TEXORA · LECTURE SHEET

Yarn and Fibre Structure and Their Properties

Fabric mechanics starts one level below the fabric: with the polymer inside the fibre, the fibre inside the yarn, and the packing, twist and migration that decide how much of the fibre's strength and extension the yarn — and later the cloth — can actually use.

FM-4203 · FABRIC MECHANICS · TOPIC 1 · FABRIC MANUFACTURING SPECIALIZATION

1. The structural hierarchy

Every textile structure is built in levels, and mechanics must be studied level by level because each level modifies the properties of the one below:

- **Molecular level** — the polymer chains: their chemistry, degree of polymerisation, and the hydrogen or covalent bonds between them.
- **Supermolecular level** — how the chains pack: **crystalline regions** (ordered, chains aligned, strong, stiff, low extension) and **amorphous regions** (disordered, the source of extension, dye uptake and moisture absorption), plus the **orientation** of both toward the fibre axis.
- **Morphological level** — the fibre as an object: its cross-section shape, surface structure (cotton's convolutions, wool's scales, the trilobal or round sections of extruded filaments), its length and its fineness.
- **Yarn level** — fibres assembled with twist into a staple yarn, or filaments assembled (flat, twisted or textured) into a filament yarn.
- **Fabric level** — yarns interlaced or interlooped; the subject of Topics 3 and 4.

The working principle of the whole course: **a fabric property is a fibre property filtered through two structures**. The filter at each level is geometry — angles, packing fractions, contacts — and geometry can be calculated. That is what fabric mechanics is for.

2. Fibre properties that the fabric inherits

FIBRE	DENSITY (G/CM³)	TENACITY (CN/TEX)	BREAKING EXTENSION (%)	MOISTURE REGAIN (%)	STRUCTURE NOTE

Cotton	1.52	20–45	5–10	7–8.5	Convoluted ribbon; ~70% crystalline cellulose
Wool	1.30	9–18	25–40	13–16	Scaly surface, bilateral cortex, crimped
Viscose	1.52	20–26	15–25	11–13	Serrated section, low crystallinity — weakens wet
Polyester (PET)	1.38	35–60	15–30	0.4	Round section, highly drawn and oriented
Nylon 6.6	1.14	40–70	15–25	4–4.5	Round section, high work of rupture

Three of these columns do mechanical work later in the course. **Tenacity and breaking extension** set the ceiling for yarn and fabric strength and the shape of every load–extension curve. **Density** converts between the mass measures the spinner uses (tex) and the areas and diameters the geometer needs (Topic 2). **Regain** matters because absorbed water plasticises the amorphous regions — cotton strengthens slightly when wet, viscose loses up to half its strength, polyester is unchanged — and because a fabric's dimensions and mass are quoted at a stated atmosphere for exactly this reason.

3. Fibre fineness and the diameter conversion

Fibre fineness is specified as linear density (dtex), but geometry needs a diameter. For a fibre of linear density t (tex) and density ρ (g/cm³), the cross-sectional area is $A = t \times 10^{-3} \div \rho$ mm², and for a circular section $d = \sqrt{(4A/\pi)}$.

WORKED EXAMPLE — DIAMETER OF A COTTON FIBRE

Cotton fibre of 1.7 dtex = 0.17 tex, $\rho = 1.52$ g/cm³.

$A = 0.17 \times 10^{-3} \div 1.52 = 1.12 \times 10^{-4}$ mm².

$d = \sqrt{(4 \times 1.12 \times 10^{-4} \div \pi)} = \sqrt{(1.42 \times 10^{-4})} \approx 0.0119$ mm ≈ **12 µm**.

Cotton is in fact a collapsed ribbon, not a circle, so 12 µm is its *equivalent* diameter — the diameter of the circle of equal area. Equivalent diameters are the honest currency of textile geometry: they conserve what the calculation actually uses (area, mass) while discarding shape detail the model cannot carry.

Fibre length is the second size variable. In a staple yarn, load passes from fibre to fibre by friction over the overlap length; below a critical staple length the yarn fails by fibres slipping rather than breaking, and no amount of twist fully repairs it. Length and fineness together fix the third key quantity — how many fibres share the yarn's cross-section.

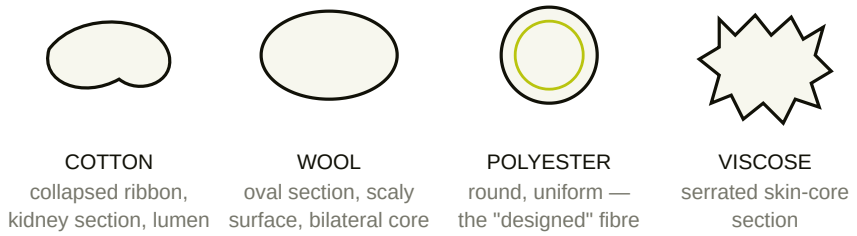


Figure 1 — Fibre cross-sections (schematic, not to a common scale): shape is a property. Cotton's collapsed ribbon convolutes and coheres in spinning; wool's scales drive felting and its bilateral cortex drives crimp; the extruded fibres carry whatever section the spinneret was cut to — serrations on viscose raise its surface area and dye uptake.

4. How many fibres in the yarn cross-section?

$$n = \text{yarn linear density (tex)} \div \text{fibre linear density (tex)}$$

WORKED EXAMPLE — FIBRES PER CROSS-SECTION

A 20 tex cotton yarn spun from 1.7 dtex (0.17 tex) fibre:

$n = 20 \div 0.17 \approx$ **118 fibres**.

Ring spinning becomes irregular and weak when n falls much below ~50–80 for cotton (the count of fibres is a statistical population: thin places are places where the population randomly thinned). The same 20 tex in a 3.3 dtex fibre would carry only ~61 fibres — coarser fibre, same count, a measurably less even and usually weaker yarn. This single ratio explains most of the practical link between fibre fineness and the finest count spinnable.

5. Yarn structure: the staple yarn as an assembly

A staple yarn is not a solid rod but a **helical assembly of finite fibres**, and its structure has four features that Topic 2 will quantify:

- **Twist:** the fibres follow helices around the yarn axis. Twist generates the inward pressure that makes fibres grip; it also tilts fibres away from the loading direction, wasting some of their strength. Yarn strength versus twist is therefore a curve with an optimum, not a staircase.
- **Packing:** fibres occupy only a fraction of the yarn's envelope — the **packing fraction**, typically 0.55–0.65 for ring-spun cotton — the rest is air. Packing rises with twist and with fibre fineness

(fine fibres pack closer).

- **Migration:** fibres do not stay at one radius; during spinning each fibre migrates between core and surface along its length. Migration locks the structure together (a fibre held at many radii cannot slip out whole) and is a principal reason a staple yarn holds together at all.
- **Hairiness:** fibre ends and loops protruding from the surface — a structural defect for weaving (friction, entanglement between ends) and a feature for handle and thermal insulation.

Filament yarns remove the length problem (every filament is continuous) but keep the geometry problem: twist or interlacing still controls cohesion, and texturing deliberately destroys the straight filament geometry to gain bulk and extension. In both cases the mechanical questions of Topic 2 — what diameter, at what helix angle, with what packing — are the questions that decide fabric behaviour.

Glossary

Crystalline / amorphous regions — ordered and disordered chain packing within a fibre. **Orientation** — alignment of chains/crystallites toward the fibre axis. **Equivalent diameter** — diameter of the circle of equal cross-sectional area. **Packing fraction** — fibre volume ÷ yarn envelope volume. **Migration** — the radial wandering of a fibre along the yarn length. **Tenacity** — breaking load per unit linear density (cN/tex).

Key points

• Properties flow upward: fibre → yarn → fabric, filtered at each level by geometry.

• Diameter from linear density: $A = t \times 10^{-3}/\rho$ mm², $d = \sqrt{(4A/\pi)}$ — a 1.7 dtex cotton fibre ≈ 12 µm equivalent diameter.

• Fibres per cross-section n = yarn tex ÷ fibre tex (worked: 118) governs evenness and the finest spinnable count.

• A staple yarn is a helical, migrating, partly-air assembly: packing fraction ~0.55–0.65, and twist both creates and wastes strength.

Practice

1. Find the equivalent diameter of a polyester fibre of 1.4 dtex (ρ = 1.38 g/cm³).
2. How many fibres occupy the cross-section of a 24 tex yarn spun from 1.5 dtex fibre? Comment on spinnability.
3. Explain why viscose fabric loses strength when wet while polyester fabric does not, in terms of supermolecular structure.
4. State two distinct mechanical consequences of fibre migration in a staple yarn.

Answer key

1. $A = 0.14 \times 10^{-3} \div 1.38 = 1.01 \times 10^{-4}$ mm²; $d = \sqrt{(4 \times 1.01 \times 10^{-4} \div \pi)} \approx 0.0114$ mm ≈ **11 µm**.
2. n = 24 ÷ 0.15 = **160 fibres** — a generous population; evenness should be good and the count is comfortably spinnable.
3. Water enters and plasticises the **amorphous** regions, breaking hydrogen bonds between chains. Viscose has a large amorphous fraction and low crystallinity, so much of its strength is bond-based and is lost wet; polyester has very low regain (0.4%) — water barely enters the structure, so its strength is essentially unchanged.
4. (i) **Cohesion:** a migrated fibre is gripped at many radii along its length, so it cannot withdraw whole — slippage resistance and yarn strength rise. (ii) **Uniformity of stress:** migration continually exchanges fibres between the highly strained surface and the slack core, evening out load sharing (and it also produces the surface hairs when fibre ends are the migrating part).
